

Md Mazharul Islam

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Authorized to work in Canada for any employer

- Software engineer with 6 years building and operating distributed backend services, most recently at Microsoft
- Built high-availability identity and security systems at consumer scale, owning services end to end including on-call
- Production cloud experience spanning distributed data stores, event streaming, caching, CI/CD and observability

SKILLS

Languages: Python, Java, Scala, C#, JavaScript, TypeScript, C++

Cloud & Backend (Azure/AWS): Azure DevOps, Azure Synapse Pipeline, PySpark, Azure Cosmos DB, Azure Cache for Redis, Azure Monitor, Azure Managed Grafana, Azure Event Hubs, AWS (S3, EC2), RESTful APIs

Frameworks & Tools: Spring Boot, .NET, Flask, Django, React, Next.js, MySQL, Docker, Git, Pandas

PROFESSIONAL EXPERIENCE

Software Engineer, Microsoft Corporation, Vancouver, Canada (*Internal Relocation*) Jun 2023 – Feb 2025

- Contributed to Microsoft's company-wide Secure Future Initiative to safeguard customer data from bad actors
- Migrated 7 Identity Protection services from key/certificate-based to Managed Service Identity (MSI) authentication
- Built an Azure Synapse pipeline (PySpark) validating ~144M IP geolocation records daily
- Cut ~2 hours from each deployment cycle by automating environment configuration and region-wise rollouts for bi-weekly releases across 7 regions and 50+ VMs (Azure DevOps, PilotFish)
- Wrote TSG for redirecting traffic to alternate regions during incidents, adopted by on-call engineers for faster recovery
- Onboarded 3 new hires, ramping them on service architecture and on-call responsibilities

Software Engineer, Microsoft Corporation, Redmond, USA Jul 2022 – Jun 2023

- Developed and maintained Identity Protection Services features serving 42.6M+ customers at 99.99% availability
- Reduced abnormal empty IP location detection time from 120 to 45 mins by creating Geneva metrics and updating monitor logic
- Developed Live Site Manager (LSM) jobs in Geneva Actions to expedite data uploads and runtime config changes, reducing issue mitigation time by ~70% (16 hrs to 4.5 hrs) and limiting customer impact during live site incidents
- Delivered 4 repair items in a cross-team effort to harden service resiliency after a Sev 1 incident

Implementation Consultant (*Software Development*), Edifecs, Bellevue, USA May 2021 – Jun 2022

- Built validation, transformation and conversion features in XEngine, Edifecs' transaction processing engine
- Developed APIs and DRL rules for Encounter Management of insurance claims, deploying through Azure Pipelines CI/CD

Graduate Research Assistant, MuSyC Lab, Missouri State University Aug 2018 – Apr 2021

- Built a source code recommendation tool (Python, Flask) with query reformulation requiring no crowd-sourced metadata
- Developed a source code summarizer using an unsupervised natural language processing method
- **Publication:** SoCeR: A New Source Code Recommendation Technique for Code Reuse (IEEE COMPSAC 2020)

Software Engineer, Synesis IT Limited, Bangladesh May 2016 – Jul 2018

- Prototyped ~250M bio-metric records into a centralized API system for SIM operations, ensuring 500 TPS via PL/SQL stored procedures with Spring JPA, Hibernate, and REST APIs
- Developed management software for ~10k employees, migrating ~110M population census records from a legacy system using .NET and Entity Framework

PROJECTS

Interview Coach, Personal Project Jun 2025 – Aug 2025

- Built and deployed a full-stack AI-powered system-design interview practice platform (Next.js, TypeScript, Supabase, Groq LLM) delivering multi-turn conversational evaluation with structured, rubric-validated feedback
- Designed rate limiting at three levels (edge, per session, and a daily budget) to keep running costs near zero
- Deployed on Vercel and verified the live behavior, including state persistence and rate limiting under real traffic

EDUCATION

Missouri State University, Springfield, USA Aug 2018 – May 2020

Masters in Computer Science

Bangladesh University of Engineering & Technology, Bangladesh Feb 2011 – Mar 2016

Bachelor of Science in Computer Science and Engineering